TRA

TRANSPORT RESEARCH ARENA BUDAPEST

18-21/05/26

Responsible AI Agents in Transportation Research

Opportunities and Risks for Ethical and Impactful Innovation

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Presentation Framing

This presentation is about:

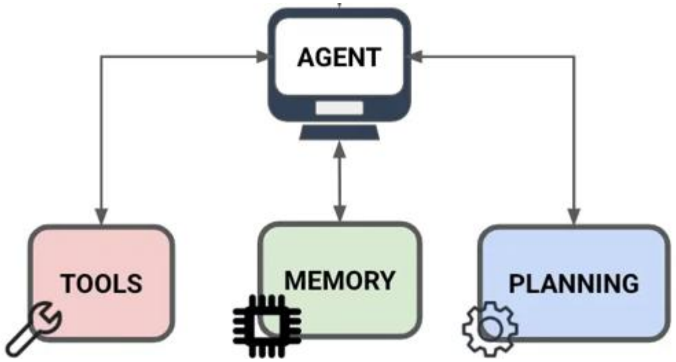
AI agents

not Artificial Intelligence

Transportation Research Process

not Research Outputs

AI agent is an autonomous or semi-autonomous system that can **plan, reason, and execute** multi-step tasks to assist with or conduct scientific inquiry.



Research Process is a systematic process for identifying transportation problems, collecting and analyzing relevant data, and translating findings into evidence-based conclusions and recommendations.

Presentation Outline

1 **AI Agents in Research**
   - Why transportation is the perfect domain?

2 **Responsible AI**
   - Principles, risks, and two-way oversight

3 **Responsible AI Agent Examples**
   - Compliance officer, bias auditor, and equitable optimization

4 **Final remarks and Open Challenges**
   - Entry points and directions

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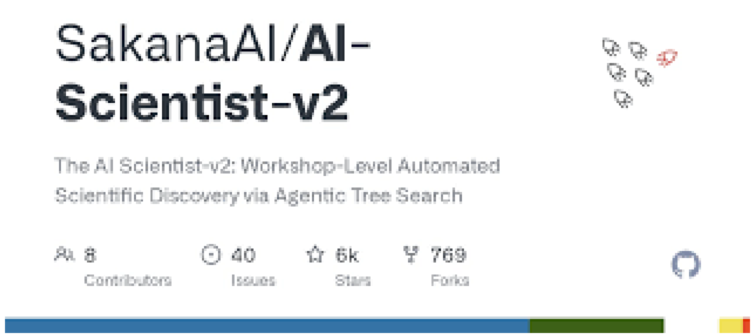
Presentation Repository

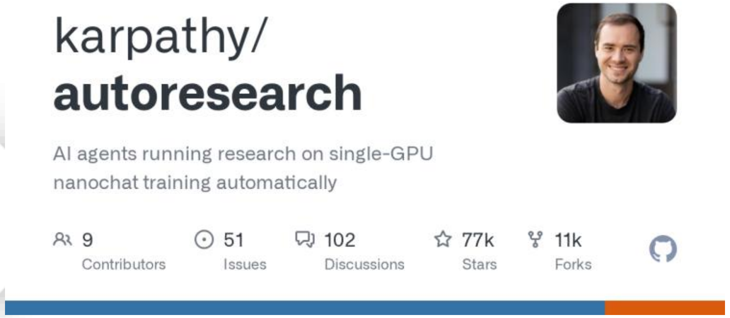
AI Agents

in Research

AI Agents in Research

The landscape of AI research agents is evolving rapidly:





SakanaAI AI-Scientist-v2
Introduced: Aug 2024

Success story:
accepted at the ICLR workshop;
ALE-Agent's 1st place in optimization

Google Co-Scientist
Introduced: Feb 2025

Success story:
antimicrobial resistance in ICL
Untapped frontier for transportation

Karpathy's Autoresearch
Introduced: Mar 2026

Success story:
Shopify model
Untapped frontier for transportation

Opportunity: AI Agent in Research

karpathy/
autoresearch

AI agent to run machine learning experiments automatically: implement a model, train, evaluate.

- **Discover better architectures** that outperform state-of-the-art models for network-wide traffic prediction.
- **Automate feature engineering** – incorporating specific weather datasets or special event schedules

Example autoresearch programme for urban traffic forecasting

programme.md ×

C: > Users > Dmitry.Pavlyuk > OneDrive - Transporta un sakaru institūts > programme.md > # Urban traffic speed prediction

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# Urban traffic speed prediction

## Task
Predict 15-minute average vehicle speed (km/h) across 50 sensors in a city grid.
Dataset: PeMS-BAY (publicly available). Input: last 12 timesteps (3h). Output: next 12.

## Current baseline
- Model: 2-layer GRU, hidden_dim=64, lr=3e-4
- Metric: val_MAE (lower is better)
- Baseline val_MAE: ~3.8 km/h

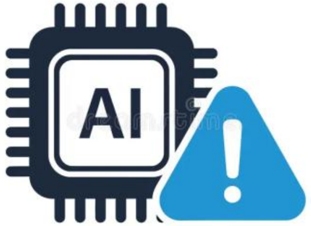
## Things to explore (try these roughly in order)
1. Temporal patterns: add hour-of-day and day-of-week embeddings (rush hour matters a lot)
2. Spatial: try a simple adjacency-weighted average of neighboring sensors as an extra input feature
3. Architecture: test LSTM vs GRU vs 1D-CNN-then-GRU
4. Loss: try Huber loss instead of MAE (robust to traffic incident outliers)
5. Input normalization: z-score per sensor vs global normalization
6. Sequence length: try 6-step (1.5h) and 24-step (6h) inputs
7. Dropout: 0.1, 0.2, 0.3 on the recurrent layer

## Constraints
- Keep each run under 10 minutes on a single GPU
- Never change the train/val/test split
- Metric is always val_MAE on the held-out val set
```

Research is shifting from writing training code to writing research directions

→ from marginal improvements to foundational research questions

AI Agents in the Research: Risks



- Hallucination
  - **AI agents may fabricate results or misinterpret data**
- Reproducibility level
  - **AI pipelines make it hard to verify findings**
- Over-reliance and skill atrophy
  - **Researchers may lose methodological skills**
- Bias amplification
  - **AI pipelines can replicate historical biases in transport datasets**
- Authorship ambiguity
  - **AI agent-generated discoveries raises ethical and legal issues**

But aren't these problems already present in traditional research?

AI Agents in Transportation Research

Why transportation is the perfect domain for AI agents?

1 **Transportation data is an ideal fuel for AI agents**
   - sensor data, probe car data, crash records, emissions logs

2 **Perfect match with AI advantages**
   - AI agents are at the expert level of transportation research tools: optimization and ML training

3 **Regulatory pressure**
   - GDPR, EU AI Act, standards

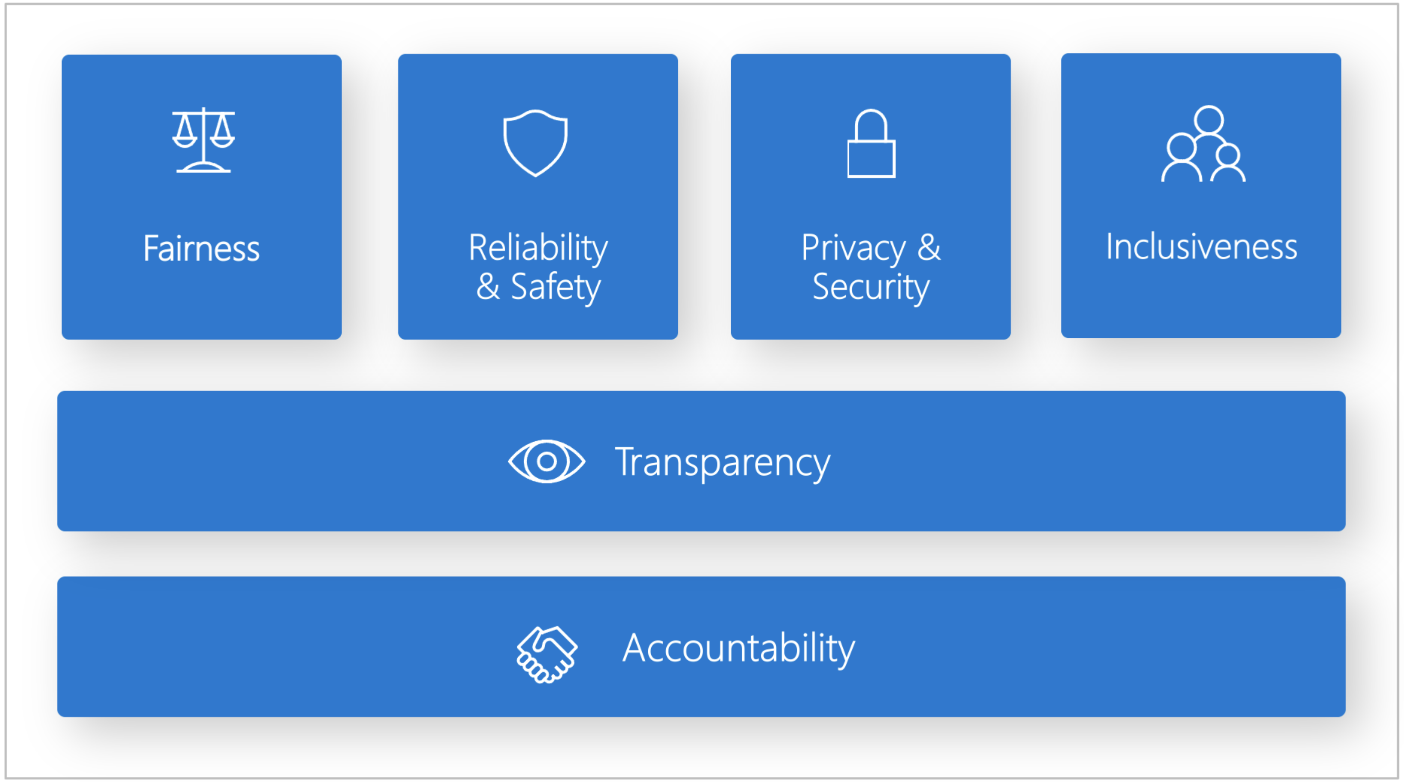
4 **Significant benefits**
   - Research acceleration or automation, AI agent’s hours vs. human’s months

Responsible AI agents

in Transportation Research

Responsible AI: Principles

Responsible AI is the practice of developing and using AI systems in a way that benefits society while minimizing the risk of negative consequences (ISO, 2023).



NIST AI RMF

AI Risk Management Framework
Jan 2023

ISO

ISO/IEC 42001: AI management
Dec 2023

EU Artificial Intelligence Act

EU AI Act
Aug 2024

Responsible AI: Risks and Mitigation

Transparency

Risk: black-box decisions
Mitigation: explainable methods (SHAP, LIME, attention, rule extraction, reasoning)
10 presentations at TRA-2026

Accountability

Risk: direct harm from AI systems
Mitigation: logs, regulatory frameworks (e.g., AI sandbox), monitoring agents

6 presentations at TRA-2026

Fairness & Bias

Risk: data inherit biased patterns
Mitigation: fairness constraints, synthetic data augmentation

5 presentations at TRA-2026

Safety & Reliability

Risk: AI must fail safely
Mitigation: human-in-the-loop, simulation-based stress tests, constrained learning
10 presentations at TRA-2026

Privacy

Risk: AI reveals sensitive information
Mitigation: federated learning, differential privacy, homomorphic encryption
5 presentations at TRA-2026

Inclusivity

Risk: Groups are left out
Mitigation: inclusive datasets, community co-design

4 presentations at TRA-2026

Responsibility Oversight is a Two-way Street

- Researchers oversee AI agents
  - Transparency
  - Accountability
  - Fairness
  - Safety
  - Privacy
  - Inclusivity
- AI agents oversee the Researchers
  - Compliance
  - Bias prevention
  - Consistency

Opportunity 1: AI Agent as a Responsible Compliance Officer

Challenge: compliance with legal and ethical rules in large-scale EU-level research projects

- Large consortiums
- Various data sources

Case: **Bio-Intel-Mob project** (Horizon Europe, 101239638)

- 32 partners across 13 countries
- 8 pilots
- Hundreds (!) of data sources, including Interviews, surveys, sensors, cameras, etc.
- 3 roles per pilot: Local Ethics Responsible, Local Data Protection Officer, Local Data steward

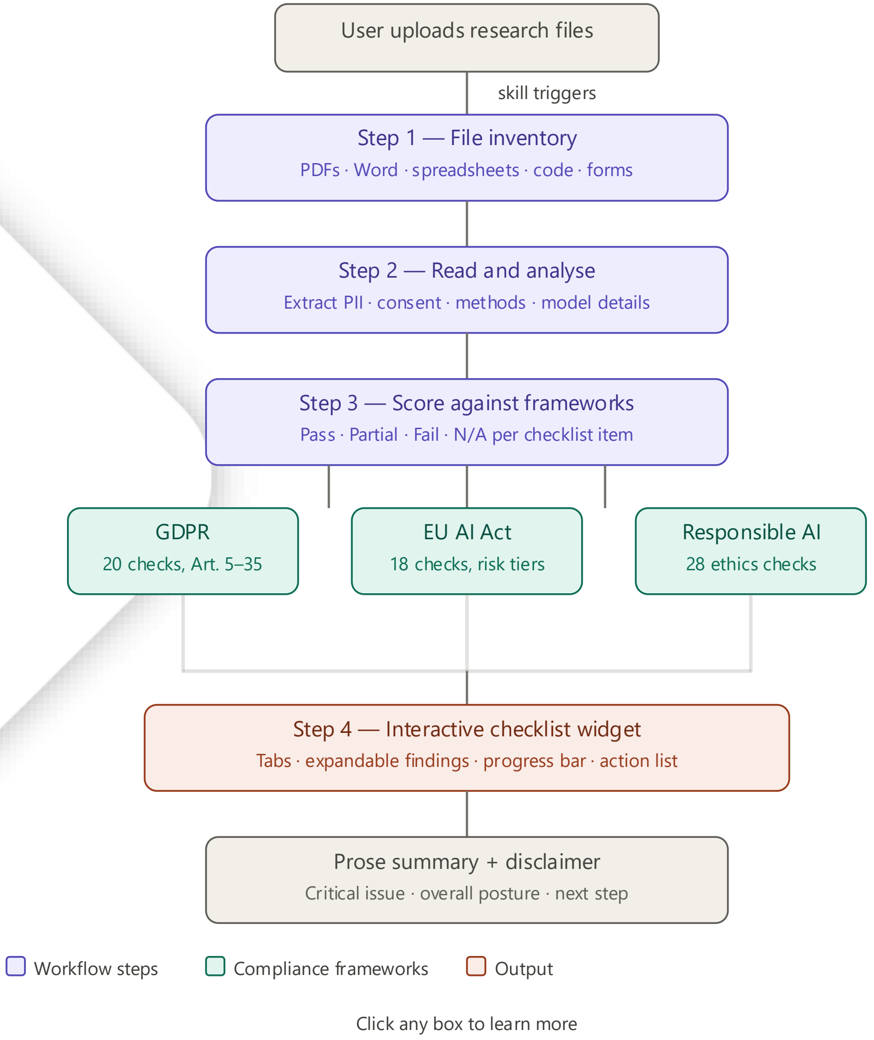
Opportunity: AI as a Responsible Compliance Officer

Opportunity 1: AI Agent as a Responsible Compliance Officer

AI Agent as a Responsible Compliance Officer:

- Connected to the Project repository
- Continuously monitors data and research protocols
- Updated with recent standards and legislation
- Flag potential issues

There are commercial solutions,
but a simple **skill** is working –
available at the presentation repo.

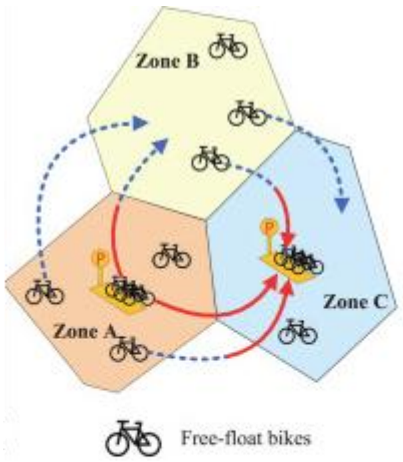


Opportunity 2: AI Agent for Penalization of Automated Decisions

Challenge: Decisions in automated/autonomous systems in are not always responsible

Example: **Empty logistics** (container, ride-hailing, e-bike rebalancing)

- Focused on demand matching and operational efficiency
- Low priority for peripheral, low-income, or minority zones
- Non explainable decisions



Fu et al. (2025)

Opportunity: an AI agent that contributes to the optimization objective

Opportunity 2: AI Agent for Penalization of Automated Decisions

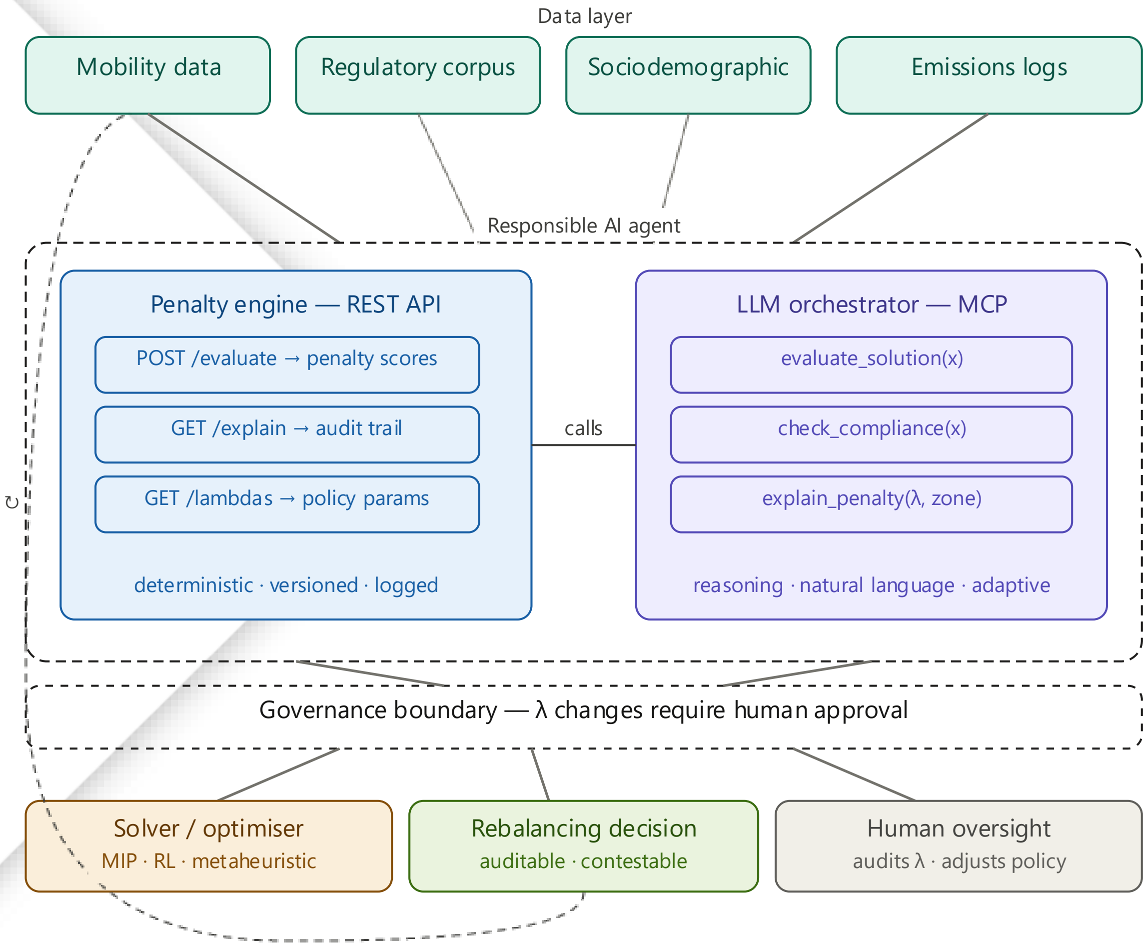
Recent publications

- *Horton et al. (2024) A scalable optimization approach for equitable facility location: Methodology and transportation applications*
- *Guo et al. (2024) Fairness-enhancing vehicle rebalancing in the ride-hailing system*

The responsible restatement:

$$\begin{aligned}\min\ & \mathrm{RepositioningCost}(x) \\ & + \lambda_1 \cdot \mathrm{EquityPenalty}(x) \\ & + \lambda_2 \cdot \mathrm{CarbonPenalty}(x) \\ & + \lambda_3 \cdot \mathrm{DriverFairnessPenalty}(x) \\ & + \lambda_4 \cdot \mathrm{AccessibilityPenalty}(x)\end{aligned}$$

where λ are a transparent, regulator-set policy parameter.



Opportunity 3: AI Agent as a Researcher's Bias Auditor

Challenge: traditional research suffers from cognitive biases

- Confirmation bias, optimism bias, anchoring bias

Case: **Cascade of confirming literature reviews**

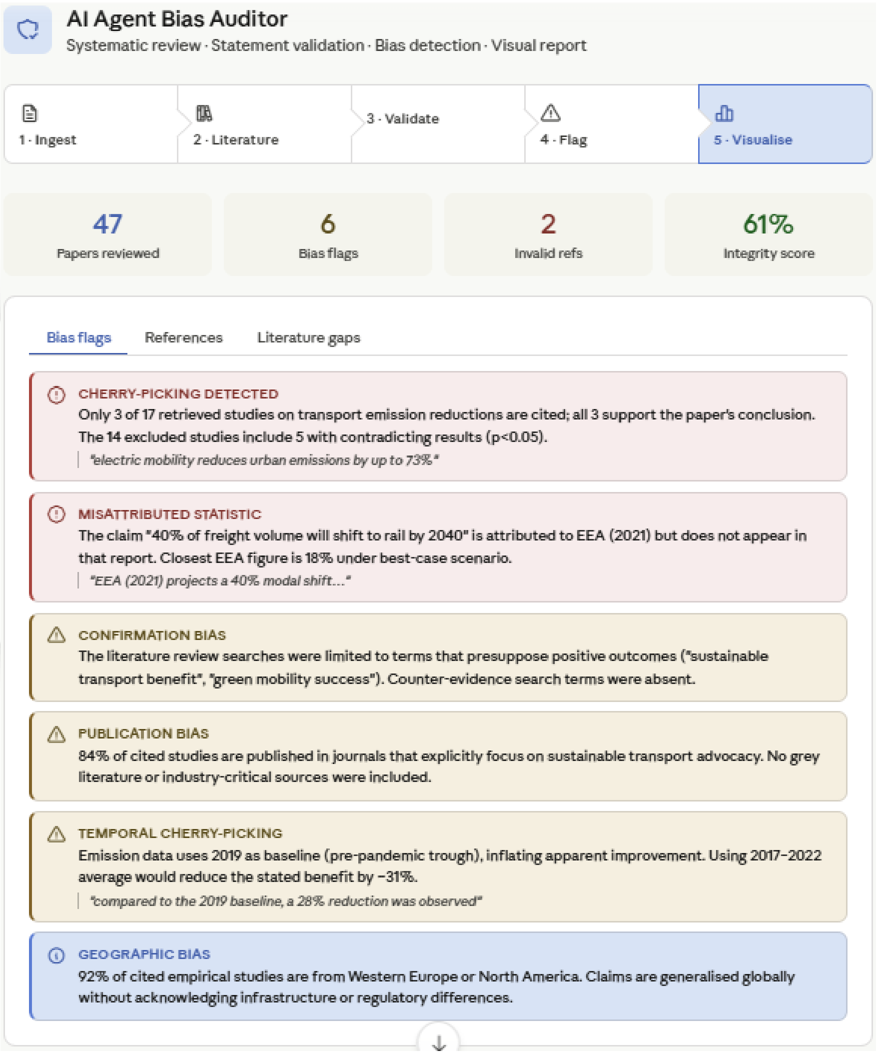
- Flyvbjerg (2002) **assumed** optimism bias in rail infrastructure projects
  9 of 10 projects overestimate demand, +106%
- Flyvbjerg (2002)
  ⇨ cited by Flyvbjerg et al. (2003, 2004, 2005...)
  ⇨ cited by Cantarelli et al. (2010, 2012...)
  ⇨ cited by Country replication studies (Dutch, Italian, Norwegian...)
  ⇨ cited by Policy documents (UK Dept. for Transport, World Bank...)
  ⇨ cited by new researchers

Opportunity: AI as a Researcher's Bias Auditor

Opportunity 3: AI Agent as a Researcher's Bias Auditor

AI Agent as Researcher's Bias Auditor:

- Run a systematic literature review
- Analyze research outputs (papers, report, deliverables)
- Validate statements and references
- Flag biases and cherry-picking



Example AI agent is available at the **presentation repository**.

Final Remark

Final Remark

What we showed

- Transportation is an ideal yet still untapped domain for AI agents
- Responsibility is not a constraint on AI agents – it is a two-way street
- Perspective patterns: AI compliance officers, bias auditor, and AI-controlled optimization

Emerging steps

- Acceptance and informal standards for AI agent use in transportation
- Reproducible, peer-reviewed transportation research generated autonomously by AI agents

Presentation Repository

Thank you!

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Re-Generation in transport

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